Thesis Proposal
Using Statistical Classification for High-dimensional Database Indexing and Information Retrieval

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1 Introduction

In advanced information systems such as a multimedia database, the content is increasingly complex; consisting of not just text but also images and audio soundtracks. Information retrieval that relies on just a few attributes is insufficient, hence content-based retrieval is required. Most systems represent objects in a database by a set of feature vectors such as color or keyword histograms, shapes descriptors and so on. When users query the database, they are usually interested retrieving similar objects. Similarity is typically measured by a distance function between feature vectors. Commonly used functions includes the Manhattan (L_1), the Euclidean (L_2) and the Maximum (L_∞) distance. These functions can also be weighted to personalize the searches. A fast and efficient method is required to search the database and to return accurate retrieval results that best match what a user has in mind. Both requisites can be satisfied by classification methods. We use classification methods for: (1) indexing a database and (2) for information retrieval.

Indexing structures can be divided into two broad categories: data-partitioning structures such as R-tree, and space-partitioning structures such as hashing, grid-files and etc. As most multimedia content is characterized by feature vectors of high dimensions, traditional indexing structures fail as they succumb to the "curse fo dimensionality". As dimensionality increases, the most obvious effect is the exponential growth of the data space volume. Most index partitions will span the most of the dimensions resulting in coarse partitions. Even then, 'dead spaces' where the partitions contain no objects are ubiquitous. The search space also grows rapidly with dimensions, sometimes encompassing the entire dataspace. More details on the effects of high-dimensionality can be gleaned from [15].

We propose the use of a clustering/classification approach to overcome the problem of high-dimensionality. We perform non-supervised clustering on the objects within the database, and thus grouping objects 'close' to each other together on disk. We model a similarity search as a classification problem. Given a query, we predict the cluster that it belongs to and search for nearest

neighbors within that cluster. In addition, we can store information about nearby clusters so that we can expand the search to the nearby neighborhood. Our approach can support approximate similarity searches at a low cost in terms of disk access, and also support dynamic feature weighting that is used in personalizing a similarity search.

We also use classification for semi-automatic image annotations so that accurate information retrieval can be realized. A user can pose a simple query such as "find images of bears". Once the keywords are extracted from the query, we can use our classification scheme to identify the categories of images that may contain images that the user wants. Because we make use of ensemble classifiers, we are able to classify an image into more than one category (fuzzy classification). This allows us to support queries such as "find images of tiger swimming in a river" where we return images that are highly-related to tigers and water.

The rest of the proposal is organized as follows. Section /refsec-related discusses related work in indexing and classification. Section /refsec-res presents some preliminary results as well as highlights the open research issues. Section /refsec-outline lays out the tentative thesis structure and finally Section /refsec-goal state the research goals and milestones.

2 Related Work

We discuss the major works that has been done in the area of high-dimensional space indexing, and classification schemes. Related work for image annotation can be found in [12].

2.1 Indexing Structures

In a multimedia database, we are mostly interested in conducting nearest neighbor (NN) queries. Most existing indexing techniques are some variants on B^+-tree-like structures. The structure divides the data space into a number of subregions, each containing a subset of objects in the database. The subregions can be modeled as minimum bounding rectangles (R-tree and X-tree), or bounding spheres (SS-tree and TV-tree). There are two major techniques that have been proposed to process NN queries: Roussopoulous et al. [11] propose a "branch and bound" algorithm, while Hjaltason and Samet [9] propose using a priority queue scheme to prune the search space. Other schemes that supports approximate NN search are locality-sensitive hashing proposed by Indyk and Motwani [10], VA-file by Weber et al. [15]. Still some others attempt to index the metric distance objects instead of the feature vectors (vantage point tree, generalized hyperplane tree, M-tree) Further details on index structures related work can be found in [8] and [2].

2.2 Classification Schemes

For a $K-$class classification problem, it is computationally expensive ($O(K^3)$) to consider all classes simultaneously. A common approach is to first decompose the problem into a number of binary-class sub-problems. Then, an ensemble scheme is used to combine the solutions of these sub-problems. Representative schemes are *one per class*, *pairwise coupling*, and more recently *error-correcting output coding* [6]. Ensemble schemes can be characterized by their design goals. The first goal is to reduce classification variance caused by the training data. Some well-known methods are bagging [3], arcing [4] and boosting [13]. The second goal tries to reduce classification error by employing decomposition and reconstruction methods with error-correction capability [1, 6, 14].

3 Preliminary Results

We have perform experiments using classification for images described by high-dimensional feature vectors and using classification to perform similarity searches on image data indexed by clusters.

3.1 Image Classification

In [7], we use ensemble schemes to combine the classification results from Support Vector Machine binary classifiers to provide a unified class prediction for an object. Several ensemble schemes were evaluated:

- **One-per-class** (OPC). One SVM classifier is trained for each class, the largest classifier output determines of the object's class.
- **Pairwise coupling** (PWC). For K classes, we train $K(K-1)$ classifiers, each separating class i from class j were $i, j = 1 \ldots K$ and $i \neq j$. The class prediction is obtained by combining the classifiers' output using some decision functions.
- **Correcting classifiers** (CC). This scheme trains classifiers that separate two classes i and j from the remaining $(K-2)$ classes.
- **PWC with CC**. We add the CC classifiers' output to PWC's to reduce the effect of irrelevant classifiers. Before we want to use a PWC classifier to determine if an object belongs to class i or j, we make sure that it has a high probability of being in the either class.
- **Error-correction output coding** (ECOC). Codewords of length L are designed for each class. They should have good row and column separation so that the error-correction capability is maximized. Then, L SVM classifiers are trained. To classify an object, the SVM outputs are combined to form its codeword. This codeword is then compared with those designed for each classes and the one which gives the lowest Hamming distance determines the class.

The insight we gained runs counter-intuitive to conventional wisdom on why ensembles are effective. It was previously thought that ensembles that have good error correction capability will yield higher classification accuracy. However, we find that even a simple OPC scheme with no error-correction capability consistently outperforms ECOC. We realize that the supremacy of an ensemble scheme is related more its ability to maximize margins (defined as the probability of the true class minus the probability of the most popular wrong class) than to anything else.

Open Issues

These includes:

- Exploring methods such as bagging and boosting to strengthen the base SVM classifiers.
- Designing new ensemble schemes that can maximize margins.
- Investigate novel techniques to further improve class prediction accuracy.

3.2 Approximate Similarity Searches in High-dimensional Spaces

In [8], we implemented a clustering/classification scheme to index image data to support approximate similarity searches. We first use Tree-Structured Vector Quantization (TSVQ) to perform non-supervised clustering on the image data. Each cluster is stored in a sequential to minimize disk access overhead. A similarity search is treated as a classification problem and each cluster is

synonymous to a class. If the query object can possibly be classified into C classes, then it is highly likely that its nearest neighbors are found in those C classes.

We experimented with a few schemes to choose the clusters to which a query object can belong to. So far, using cluster centroid to yields the highest $k-$NN recall. This is not surprising as the clusters are formed by using the Euclidean distance between points to determine their closeness. We also use SVM classification and find that it works very well for $1-$NN searches. We then attempt to improve recall by employing classification error-reduction techniques such as bagging. Recall improves slightly but at the expense of increased CPU time and storage space. We also show that dynamic feature weighting can be supported only slight degradation in recall.

Open Issues

There are several open issues associated with clustering and how classification can be used:

- For TSVQ, we first need to understand the effect of distortion on recall.
- How we can refine the cluster selection to improve recall. This involves developing new metrics to rank the cluster as well as alter the design of the cluster to take into account the training data and query data distribution. Exploration of other clustering techniques beside TSVQ is needed.
- A new dynamic partial distance function (DPF) discovered by Li et al. [5] is able to model human perception much better than traditional functions such as the Euclidean distance. Using DPF to index a database can possibly improves recall. Indexing such a dynamic function can be quite a challenge.

3.3 Image Annotation

We have experimented with duplicated classification where we allow an image to belongs to two classes. We find that we can improve the classification accuracy by two-fold. Instead of assigning one class to an image while annotating an image, we use the "bag of words" approach and classify each image by a vector of classes. The value of a vector element is the probability that the image belongs to that class. This is similar to a content-based dictionary. The retrieval process benefits from the fuzzy classification which gives a richer description of the image.

We did some preliminary work using CC classifiers and 14 image categories (*architecture*, *bears*, *clouds*, *elephants*, *fabrics*, *fireworks*, *flowers*, *food*, *landscape*, *people*, *textures*, *tigers*, *tools*, and *waves*). A query takes the form of "find images with x and y". We find that categories with homogeneous and distinctive features (such as tigers, bears, elephants and clouds) performs better than those with heterogeneous features (such as architecture and people).

Open Issues

Given that we were only using very general categories of images, our approach has much room for improvements. Issues that need to be addressed includes:

- How semi-automatic image annotation can be supported efficiently.
- How we can combine classifiers' output to produce an optimal fuzzy classification such that the accuracy of the retrieval results are maximized.
- Constructing a thesaurus of images that incorporates users' relevance feedback to enhance text-based searches.

4 Research Goals and Milestones

In our research, we want to accomplish the following:

1. Image annotation. We need to work out a scheme to optimize fuzzy classification. In addition, we want to enhance the retrieval with text by using a thesaurus such as Wordnet to perform query expansion. We also want to incorporate relevance feedback to personalize the retrieval. We will work on an image database of $51,000$ images that is characterized into about 100 categories. This work will be done jointly with Gerard Saychay.
Milestone: Target to complete a preliminary version and submit to ACM Multimedia on 15 April 2002.

2. Cluster ranking. Devise a cluster ranking algorithm to further improves NN recall. The immediate task involves the use of deterministic annealing on TSVQ to produce clusters that achieve a global minimum in distortion. This should gives us better clusters, meaning similar objects are more likely to fall into the same cluster. Hopefully, this can improve the NN recall. We would also like to mine other techniques used in the signal-processing domain to improve clustering and classification.
Milestone: Target for ACM SIGMOD or VLDB 2003 deadline.

3. Classification. We want to explore the use of combining bagging and boosting to improve accuracy. In addition, we want to design a better ensemble scheme that is able to maximize margins rather than error-correction ability.
Milestone: 2002-3.

4. New clustering schemes. Investigate other schemes other than besides TSVQ.
Milestone: 2002-3.

5. Indexing DPF. Construct an indexing structure using dynamic partial distance function. As the function is still a recent discovery and unlike any conventional metrics, we need to expend more thoughts on how to use it. This work will be done jointly with Beitao Li.
Milestone: 2002-3.

6. Case study. To show that our methods are practical and scalable, we want to index a major art collection and provide retrieval service to it. Working with such a collection is challenging because its large size, variety of images, many types of applications and diversity of users.
Milestone: 2004.

5 Committee Members

1. Edward Chang (Committee Chair) *Department of Computer Engineering*
2. Kwang-Ting Cheng (Committee Member) *Department of Computer Engineering*
3. Kenneth Rose (Committee Member) *Department of Electrical Engineering*
4. Yuan-Fang Wang (Committee Member) *Department of Computer Science*

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